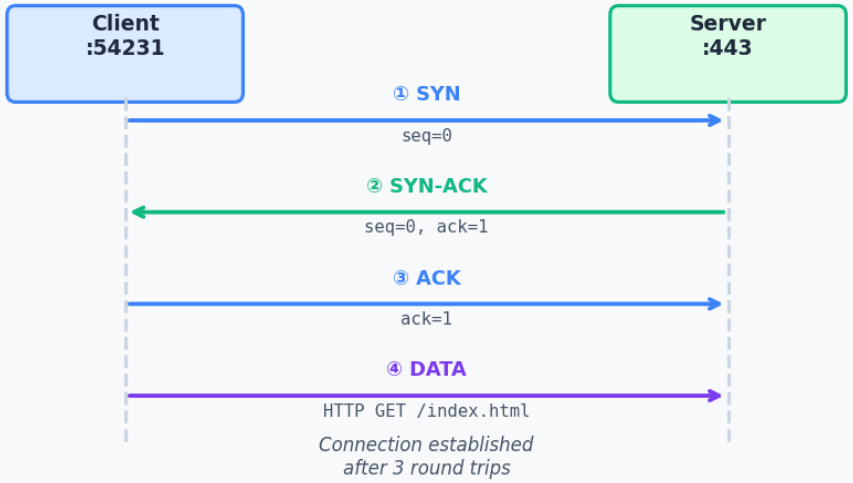
Layer 4: Transport

L4 Transport

End-to-end delivery between applications. Ports multiplex streams. TCP vs UDP is the central reliability/speed tradeoff.

QUIC (RFC 9000 / HTTP/3) Runs over UDP but reimplements reliability. Integrates TLS 1.3 (0-RTT vs TCP+TLS's 3+ round trips). No head-of-line blocking. Survives WiFi→5G handoff without dropping connections. Used by Google, YouTube, Meta for a major fraction of traffic.

TCP — 3-Way Handshake



TCP vs UDP — Header Comparison

TCP Header (20-60 bytes)

Src Port (2B)	Dst Port (2B)	Seq # (4B)	Ack # (4B)	Flags (2B)	Window (2B)	Checksum	Data...

UDP Header (8 bytes only!)

Src Port (2B)	Dst Port (2B)	Length (2B)	Checksum (2B)	Data...

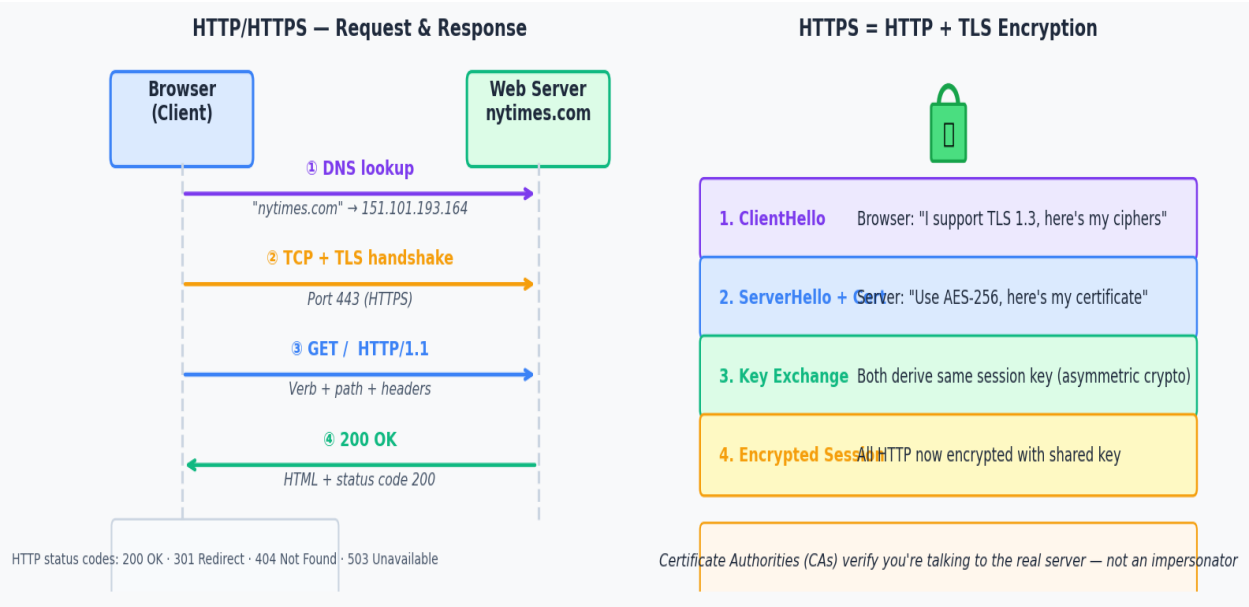
Feature	TCP	UDP
Connection	Required (handshake)	None
Reliability	Guaranteed	Best-effort
Ordering	Preserved	Not guaranteed
Speed	Slower	Faster
Use case	HTTP,SSH,SMTP,FTP	DNS,Zoom,games

KEY CONCEPT TCP trades latency for reliability. UDP trades reliability for speed. QUIC is the modern attempt to get TCP-level guarantees with UDP-level latency.

Layer 5: Application

L5 Application

Where developers work. Protocols define the language two applications speak over a network.



SMTP

Port 25 / 587

Mail transfer between servers. Plaintext by default (1980s!). STARTTLS adds encryption. Weak auth = endemic spam.

SSH

Port 22

Encrypted remote terminal. Replaced Telnet. Key-based auth. SSH tunnels forward arbitrary TCP over encrypted channel.

FTP/SFTP

Port 21 / 22

FTP (1971): plaintext incl. credentials. SFTP (over SSH): encrypted. FTP survives in legacy industrial systems.

DNS

Port 53 UDP

Hostname → IP. Hierarchical: root→TLD→authoritative. TTL caching. DoH/DoT encrypt queries. Critical infrastructure.

KEY CONCEPT Application protocols define message format, verbs, and session rules. Many were designed in the 1980s without security assumptions — which is why plaintext protocols and BGP hijacking remain live threats.

Summary: The 5-Layer Model

Every network problem has a layer. Identifying it is half the diagnosis.

5	**Application** *Software communicates*	HTTP · HTTPS · SMTP · FTP · SSH · DNS · DoH
4	**Transport** *App-to-app, reliability/speed*	TCP · UDP · QUIC · Port numbers
3	**Network** *Global routing, IP addressing*	IPv4 · IPv6 · BGP · Routers
2	**Data Link** *Local delivery, MAC addressing*	Ethernet · 802.11 MAC · ARP · DHCP · Switches
1	**Physical** *Bits → signals*	Fiber · Copper · Coax · WiFi · 5G · Microwave · Satellite

Mnemonic (bottom → top): **P**eople **D**on't **N**eed **T**o **A**sk (Physical → Data Link → Network → Transport → Application)

KEY CONCEPT The TCP/IP model is a mental framework for diagnosing failures: is this a physical issue, a routing issue, or an application protocol issue? Layer-based thinking is how professionals isolate problems fast.

Understanding how Facebook disappeared from the Internet

2021-10-04

Celso Martinho

Tom Strickx

5 min read

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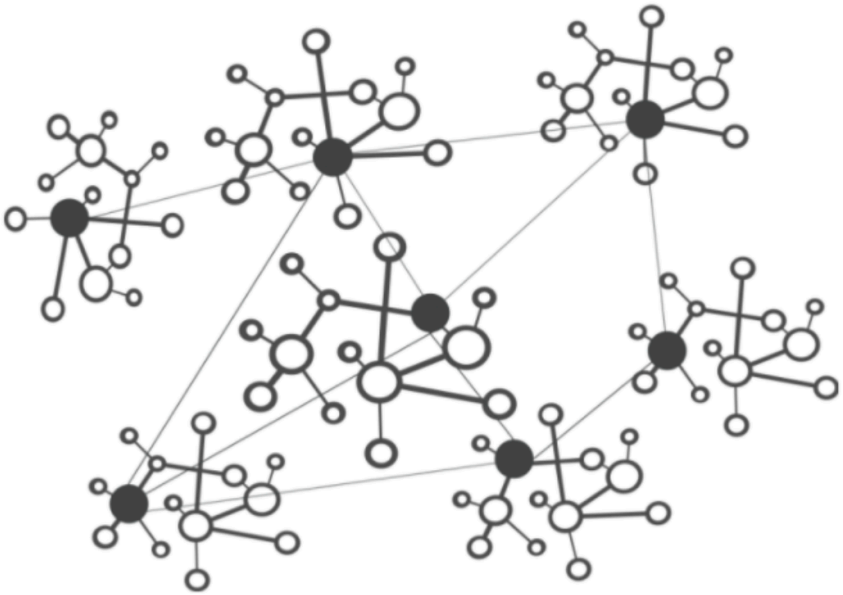

https://blog.cloudflare.com/october-2021-facebook-outage/

The Internet - A Network of Networks

"Facebook can't be down, can it?", we thought, for a second.